



GUI Application Development

Practical Skills Module 6: Topic 4

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Topic Code	PM-06-PS04
Topic	Use a known framework (of own preference) with functions to assemble a complete solution applying all the below aspects given contextualised context within a specific sector

Scope of Practical Skill

Given a range the learner must be able to:

- PA0401 Build a GUI application using Python (compulsory) which encompass any of these below:
 - Build a scientific calculator using Python
 - Digital phone book directory

Applied Knowledge

- AK0401 Concept, definition and functions of Python programming principles

Internal Assessment Criteria

- IAC0401 Expected results are achieved and the basic GUI application is operable

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-06



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Python GUI Tools

You are assigned to create two Python GUI applications for a desktop tools suite. The first is a scientific calculator capable of performing both basic and advanced mathematical operations, featuring robust input validation and error handling. The second is a digital phone book directory that enables users to add, search, update, and delete contacts.

Create a folder named **gui_projects** and complete the following:

Step 1: Build a GUI-Based Scientific Calculator

- Create a Python file called **scientific_calculator.py**.
- Design a user interface that includes:
 - A display field to show user input and calculation results.
 - Buttons for:
 - Basic arithmetic operations: addition (+), subtraction (-), multiplication (*), and division (/).
 - Advanced operations: exponentiation (^), square root (√) and trigonometric functions (sin, cos, tan).
 - Clear (C) button to reset input.
 - Equals (=) button to perform the calculation.
- Implement input validation and error handling to gracefully manage invalid cases such as division by zero and other invalid input formats.
- *Tip:* You can use PyQt5 widgets and layout classes to build the GUI, including `QApplication` and `QWidget` for the main window, `QLineEdit` for the display field,

QPushButton for the calculator buttons, QGridLayout to arrange the buttons in a grid, and QMessageBox to display error messages.

Step 2: Build a digital phone book directory using Python

- Create a Python file called **phone_book.py**.
- This program should allow users to **add, search, update, delete, and view** contact information stored in a directory.
- Build a graphical user interface that includes:
 - Input fields for contact details (e.g., name and phone number).
 - Buttons for actions such as Add, Search, Update, Delete, and View All Contacts.
 - A display area to show contacts or search results.
- Implement functionality to handle each action, storing contacts in an appropriate data structure (like a list or dictionary).
- Include input validation to prevent errors such as empty fields or invalid phone numbers.
- *Tip:* You can import the following classes from the PyQt5.QtWidgets module to build the interface: QApplication, QWidget, QVBoxLayout, QHBoxLayout, QLabel, QLineEdit, QPushButton, QListWidget, QMessageBox



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
